Computer Networks — Lecture 7 (Ch 3, Lec 1): Transport Layer — Introduction & Transport-Layer Services

Questions Bank — بنك الأسئلة

ملخص البنك: 56 سؤال اختيار من متعدد (MCQ) + 13 سؤال مقالي (Essay) + 0 سؤال حسابي (Calculation) — مفيش في المحاضرة دي. توزيع الإجابة الصحيحة في الـ MCQ: A=14 / B=14 / C=15 / D=13. الأسئلة مرتبة بحيث الـ high-likely ييجوا الأول في كل نوع.

Part 1 — Multiple Choice Questions (MCQ)

1.

Where does the transport layer reside in the layered network architecture?

ترجمة: فين طبقة النقل في الـ layered network architecture؟

A. Above the application layer

B. Between the physical layer and the data link layer

C. Between the application layer and the network layer

D. Between the network layer and the data link layer

الإجابة الصحيحة: C) Between the application layer and the network layer

ليه صح: طبقة النقل بين طبقة التطبيق (فوقها) وطبقة الشبكة (تحتها) — ده دور مركزي. مش بين physical و data link، ولا بين network و data link، ولا فوق application.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1 (intro)

Transport Layer | Architecture | ترجيح: high

.2

The transport layer provides communication services directly to:

ترجمة: طبقة النقل بتوفّر خدمات اتصال مباشرة لـ إيه؟

A. Application processes running on different hosts

B. Routers along the path

C. The physical links only

D. DNS servers only

الإجابة الصحيحة: A) Application processes running on different hosts

ليه صح: لـ application processes على hosts مختلفة. ده دورها الأساسي. مش لـ routers ولا الـ links ولا DNS.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1 (intro)

Transport Layer | Application processes | **ترجيح: high**

.3

A transport-layer protocol provides for logical communication between:

ترجمة: بروتوكول طبقة النقل بيوفّر logical communication بين إيه؟

A. Application processes running on different hosts

B. Hosts only

C. Physical links only

D. Routers only

الإجابة الصحيحة: A) Application processes running on different hosts

ليه صح: بين application processes على hosts مختلفة. من منظور التطبيق، كأن الـ hosts متصلين مباشرة رغم إنهم ممكن يكونوا على جانبي الكوكب.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1

Transport Layer | Logical communication | **ترجيح: high**

.4

By "logical communication" between processes, we mean that:

ترجمة: إيه معنى "logical communication" بين العمليات؟

A. The processes are physically connected by a wire

B. From an application's perspective, it is as if the hosts running the processes were directly connected

C. The routers handle the application messages directly

D. The transport layer routes packets through the network core

الإجابة الصحيحة: B) From an application's perspective, it is as if the hosts running the processes were directly connected

ليه صح: من منظور التطبيق، كأن الـ hosts متصلة مباشرة — رغم إن في الواقع ممكن يكونوا على جانبي الكوكب متصلين عبر راوترز وروابط.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1

Transport Layer | Logical communication | Definition | ترجيح: high

.5

On the sending side, the transport layer converts application-layer messages into:

ترجمة: على جانب الإرسال، طبقة النقل بتحوّل رسائل التطبيق لـ إيه؟

A. Network-layer datagrams

B. Transport-layer segments

C. Physical bits

D. Application frames

الإجابة الصحيحة: B) Transport-layer segments

ليه صح: لـ transport-layer segments الـ network layer بعدها بـ encapsulate الـ segment جوه datagram. مش datagrams (ده شغل الـ network layer).

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1

Transport Layer | Segments | Sending side | ترجيح: high

.6

How does the transport layer create a transport-layer segment?

ترجمة: إزاي طبقة النقل بتعمل transport-layer segment؟

A. By routing the message through the network core

B. By encrypting the entire application message

C. By compressing the message into a single byte

D. (Possibly) breaking the application messages into smaller chunks and adding a transport-layer header to each chunk

الإجابة الصحيحة: D) (Possibly) breaking the application messages into smaller chunks and adding a transport-layer header to each chunk

ليه صح: بتكسر رسالة التطبيق لـ chunks أصغيرة (لو لزم) وتضيف transport-layer header لكل chunk. ده اللي بيعمل segment. مش تشفير ولا توجيه ولا ضغط.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1

Transport Layer | Segments | Encapsulation | **ترجيح: high**

.7

The transport layer passes the segment to the network layer, where the segment is:

ترجمة: طبقة النقل بتسلم الـ segment لطبقة الشبكة، وهناك الـ segment بيتعمله إيه؟

A. Discarded and re-created at the destination

B. Encapsulated within a network-layer packet (a datagram) and sent to the destination

C. Sent directly to the application layer at the destination

D. Stored in the routers along the path

الإجابة الصحيحة: B) Encapsulated within a network-layer packet (a datagram) and sent to the destination

ليه صح: بيـ encapsulate جوه network-layer packet (datagram) ويتابع للـ destination. الـ routers بتشتغل على الـ network-layer fields فقط.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1

Transport Layer | Network layer | Encapsulation | **ترجيح: high**

.8

Network routers act on which fields of the datagram?

ترجمة: الـ routers بتشتغل على أي fields في الـ datagram؟

A. Only the network-layer fields; they do not examine the transport-layer segment fields
B. Only the transport-layer fields
C. Only the application-layer fields
D. Both the network-layer and transport-layer fields

الإجابة الصحيحة: A) Only the network-layer fields; they do not examine the transport-layer segment fields

ليه صح: على network-layer fields فقط — ما بتفحصش transport-layer segment fields. ده نقطة مهمة: الـ transport layer بتعيش في الـ end systems فقط.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1

Routers | Network layer | Transport layer | **ترجيح: high**

.9

On the receiving side, what does the network layer do with the transport-layer segment?

ترجمة: على جانب الاستقبال، طبقة الشبكة بتعمل إيه بالـ transport-layer segment؟

A. Forwards it to another router
B. Encrypts it before passing to the transport layer
C. Extracts it from the datagram and passes it up to the transport layer
D. Discards it after routing

الإجابة الصحيحة: C) Extracts it from the datagram and passes it up to the transport layer

ليه صح: بتـ extract الـ segment من الـ datagram وبتسلمه لطبقة النقل فوق. بعدها طبقة النقل بتعالجه وتسلم البيانات للتطبيق.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1

Network layer | Receiving side | **ترجيح: high**

.10

How many transport-layer protocols does the Internet make available to network applications?

ترجمة: الإنترنت بيوفّر كام بروتوكول نقل للتطبيقات؟

A. One — only TCP

B. Three — TCP, UDP, and ICMP

C. Two — TCP and UDP

D. Five

الإجابة الصحيحة: C) Two — TCP and UDP

ليه صح: بروتوكولين: TCP و UDP. كل واحد بيوفّر set مختلف من خدمات النقل للتطبيق.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1

Transport Layer | TCP | UDP | **ترجيح: high**

.11

Whereas a transport-layer protocol provides logical communication between processes, a network-layer protocol provides logical communication between:

ترجمة: بروتوكول طبقة الشبكة بيوفّر logical communication بين إيه (مقابل طبقة النقل اللي بين العمليات)؟

A. Hosts

B. Processes

C. Applications

D. Routers

الإجابة الصحيحة: A) Hosts

ليه صح: بين hosts. ده فرق أساسي: transport = process-to-process، network = host-to-host. النقطة دي subtle بس مهمة.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.1

Transport vs Network | Hosts | Processes | **ترجيح: high**

.12

The distinction between transport-layer (process-to-process) and network-layer (host-to-host) communication is:

ترجمة: الفرق بين transport layer (process-to-process) و network layer (host-to-host) إيه طبيعته؟

A. Irrelevant

B. Subtle but important

C. False — both are process-to-process

D. False — both are host-to-host

الإجابة الصحيحة: B) Subtle but important

ليه صح: subtle لكن مهم. الـ transport بيكمل شغل الـ network بيوصله من host- لـ process-process. تعريف مهم الدكتور بيشدد عليه.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.1

Transport vs Network | Distinction | **ترجيح: high**

.13

Where do transport-layer protocols live?

ترجمة: بروتوكولات طبقة النقل بتعيش فين؟

A. In the physical links

B. In the end systems

C. In the network core

D. In the routers

الإجابة الصحيحة: B) In the end systems

ليه صح: في الـ end systems. مش في الـ routers ولا في الـ network core. ده عشان كده الـ routers ما بتعترفش بمعلومات الـ transport.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.1

Transport Layer | End systems | **ترجيح: high**

.14

Within an end system, what does a transport protocol do (regarding the network edge)?

ترجمة: جوه الـ end system، بروتوكول النقل بيعمل إيه بالنسبة للـ network edge؟

A. Encrypts messages for the network layer

B. Moves messages from application processes to the network edge (the network layer) and vice versa

C. Manages the routers along the path

D. Routes packets through the network core

الإجابة الصحيحة: B) Moves messages from application processes to the network edge (the network layer) and vice versa

ليه صح: بيحرّك رسائل من العمليات لـ network edge (لـ network layer) والعكس. مالوش أي دور في إزاي الرسائل بتتحرك جوه الـ network core.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.1

Transport Layer | Network edge | **ترجيح: high**

.15

Does a transport protocol have any say about how messages are moved within the network core?

ترجمة: هل بروتوكول النقل عنده أي رأي في إزاي الرسائل بتتحرك جوه الـ network core؟

A. No, it does not have any say

B. Yes, it routes the messages itself

C. Yes, it encrypts them inside the core

D. Yes, it controls the routers

الإجابة الصحيحة: A) No, it does not have any say

ليه صح: لا — مالوش أي دور. الـ transport بتعيش في الـ end systems فقط والـ routers ما بتعترفش بمعلوماتها.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.1

Transport Layer | Network core | **ترجيح: high**

.16

Intermediate routers recognize what information that the transport layer may have added to application messages?

ترجمة: الـ routers الوسيطة بتعترف بإيه من معلومات طبقة النقل اللي ممكن تكون اتضافت؟

A. They act on it and route based on it

B. They encrypt it before forwarding

C. They neither act on nor recognize any such information

D. They recognize it but do not act on it

الإجابة الصحيحة: C) They neither act on nor recognize any such information

ليه صح: لا بتعترف ولا بتشتغل عليها. الـ routers بتشتغل على network-layer fields فقط — معلومات الـ transport ليها شأنها بس في الـ end systems.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.1

Routers | Transport Layer | **ترجيح: high**

.17

A computer network may make available how many transport protocols?

ترجمة: الشبكة ممكن توفّر كام بروتوكول نقل؟

A. None — applications handle transport themselves

B. Only one

C. Multiple, with each protocol offering a different service model to applications

D. Always exactly three

الإجابة الصحيحة: C) Multiple, with each protocol offering a different service model to applications

ليه صح: ممكن كذا بروتوكول، كل واحد بيوفّر service model مختلف للتطبيقات. الإنترنت عنده TCP و UDP كمثال.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.1

Transport Layer | Service models | **ترجيح: high**

.18

The services that a transport protocol can provide are often constrained by:

ترجمة: الخدمات اللي بروتوكول النقل يقدر يوفّرها بتبقى مقيدة غالبًا بـ إيه؟

A. The DNS resolver

B. The physical layer's bandwidth only

C. The application layer's preferences

D. The service model of the underlying network-layer protocol

الإجابة الصحيحة: D) The service model of the underlying network-layer protocol

ليه صح: بـ service model بتاع الـ network-layer protocol اللي تحته. لو الـ network ما بيوفرش delay/bandwidth guarantees، الـ transport ما يقدرش يوفرها للتطبيقات.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.1

Transport Layer | Network Layer | Constraints | **ترجيح: high**

.19

If the network-layer protocol cannot provide delay or bandwidth guarantees, what can the transport-layer protocol do?

ترجمة: لو الـ network-layer ما بيوفرش delay أو bandwidth guarantees، الـ transport-layer يقدر يعمل إيه؟

A. It can still provide them by using a different physical link
B. It can still provide them by buffering in the routers
C. It cannot provide delay or bandwidth guarantees for application messages sent between processes
D. It can still provide them by encrypting the segments

الإجابة الصحيحة: C) It cannot provide delay or bandwidth guarantees for application messages sent between processes

ليه صح: ما يقدرش يوفّرها. الـ transport بتبقى constrained بـ service model بتاع الـ network اللي تحتها.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.1

Transport Layer | Network Layer | Constraints | **ترجيح: high**

.20

What service can a transport protocol offer even when the underlying network protocol is unreliable (loses, garbles, or duplicates packets)?

ترجمة: إيه خدمة بروتوكول النقل يقدر يوفّرها حتى لو الـ network protocol مش موثوق؟

A. Lower delay than the network layer
B. Reliable data transfer service to an application
C. Encryption of all messages
D. Higher bandwidth than the network layer

الإجابة الصحيحة: B) Reliable data transfer service to an application

ليه صح: reliable data transfer service. ده مثال على إن الـ transport يقدر يضيف خدمات حتى لو الـ network تحتها ما بيوفرشها — TCP بيعمل كده فوق IP.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.1

Transport Layer | Reliable data transfer | **ترجيح: high**

.21

What can a transport protocol use to guarantee that application messages are not read by intruders, even when the network layer cannot guarantee confidentiality?

ترجمة: بماذا يمكن لبروتوكول النقل أن يضمن عدم قراءة الرسائل من قبل المتسللين، حتى لو كانت طبقة الشبكة لا تستطيع؟

A. Encryption

B. Acknowlegments

C. Sequence numbers

D. Timers

الإجابة الصحيحة: A) Encryption

ليه صح: encryption. ده موضوع Chapter 8 (network security). نقطة مهمة: الـ transport يقدر يضيف security فوق ما الـ network يوفره.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.1

Transport Layer | Encryption | Security | ترجيح: high

.22

The Internet makes how many distinct transport-layer protocols available to the application layer?

ترجمة: الإنترنت بيوفّر كام بروتوكول نقل مختلف للـ application layer؟

A. One

B. Two

C. Four

D. Three

الإجابة الصحيحة: B) Two

ليه صح: بروتوكولين: UDP و TCP. نقطة أساسية في بداية القسم 3.1.2.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.2 (intro)

Internet Transport | TCP | UDP | ترجيح: high

.23

UDP (User Datagram Protocol) provides which type of service to the invoking application?

ترجمة: UDP بيوفّر إيه نوع خدمة للتطبيق؟

A. Reliable, connection-oriented service

B. Unreliable, connection-oriented service

C. Reliable, connectionless service

D. Unreliable, connectionless service

الإجابة الصحيحة: D) Unreliable, connectionless service

ليه صح: unreliable + connectionless. عكس TCP اللي reliable + connection-oriented. تعريف أساسي.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.2

UDP Services ترجيح: high

.24

TCP (Transmission Control Protocol) provides which type of service to the invoking application?

ترجمة: TCP بيوفّر إيه نوع خدمة للتطبيق؟

A. Reliable, connection-oriented service

B. Reliable, connectionless service

C. Unreliable, connectionless service

D. Unreliable, connection-oriented service

الإجابة الصحيحة: A) Reliable, connection-oriented service

ليه صح: reliable + connection-oriented. عكس UDP. تعريف أساسي.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.2

TCP Services ترجيح: high

.25

When designing a network application, the application developer must:

ترجمة: لما تطور network application، المطور لازم يعمل إيه؟

A. Use only TCP

B. Specify one of the two transport protocols (TCP or UDP)

C. Use only UDP

D. Implement their own transport protocol

الإجابة الصحيحة: B) Specify one of the two transport protocols (TCP or UDP)

ليه صح: يحدد واحد من البروتوكولين (TCP أو UDP). ما يقدرش يكتب بروتوكول نقله الخاص، ولا هو مضطر يستخدم واحد بس.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.2

Transport Layer | Application developer | ترجيح: high

.26

In this book, the transport-layer packet is referred to as:

ترجمة: في الكتاب ده، الـ transport-layer packet بيتسمى إيه؟

A. A bit

B. A segment

C. A frame

D. A datagram

الإجابة الصحيحة: B) A segment

ليه صح: الكتاب اختار ده عشان يتجنب اللبس. RFCs بتستخدم segment للـ TCP و"datagram" للـ UDP وللـ network-layer ساعات.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.2 (terminology)

Transport Layer | Terminology | Segment | ترجيح: high

.27

The Internet's network-layer protocol is called:

ترجمة: بروتوكول طبقة الشبكة في الإنترنت بيتسمى إيه؟

A. HTTP

B. UDP

C. TCP

D. IP (Internet Protocol)

الإجابة الصحيحة: D) IP (Internet Protocol)

ليه صح: IP (Internet Protocol). TCP و UDP بروتوكولات نقل، HTTP بروتوكول تطبيق. تعريف أساسي.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.2

Network Layer | IP | **ترجيح: high**

.28

What kind of communication does IP provide?

ترجمة: IP بيوفّر إيه نوع اتصال؟

A. Logical communication between routers

B. Logical communication between hosts

C. Logical communication between processes

D. Physical communication between hosts

الإجابة الصحيحة: B) Logical communication between hosts

ليه صح: logical communication بين hosts. الـ transport layer (TCP/UDP) بيوصل ده لـ process-to-process.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.2

IP | Logical communication | Hosts | **ترجيح: high**

.29

The IP service model is best described as:

ترجمة: IP service model بيتوصف بإيه؟

A. A best-effort delivery service

B. A reliable delivery service

C. A guaranteed-bandwidth delivery service

D. A connection-oriented delivery service

الإجابة الصحيحة: A) A best-effort delivery service

ليه صح: IP بيعمل best-effort delivery service. بيوصل الـ segments، بس ما بيضمنش حاجة. تعريف أساسي.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.2

IP | Best-effort | Service model **ترجيح: high**

.30

Which of the following does IP NOT guarantee?

ترجمة: IP ما بيضمنش إيه؟

A. Segment delivery only

B. Integrity of data in segments only

C. All of these: segment delivery, orderly delivery, and integrity of data in segments

D. Orderly delivery of segments only

الإجابة الصحيحة: C) All of these: segment delivery, orderly delivery, and integrity of data in segments

ليه صح: ما بيضمنش أي حاجة من التلاتة: لا وصول، لا ترتيب، لا سلامة البيانات. عشان كده IP اسمه unreliable service. ده عكس الـ All/None of the above، إجابة "All of these" صحيحة كتفصيل هنا.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.2

IP | Unreliable | Guarantees **ترجيح: high**

.31

IP is said to be an unreliable service because:

ترجمة: ليه IP بيتوصف بأنه unreliable service؟

A. It does not guarantee segment delivery, orderly delivery, or integrity of the data in segments
B. It always loses packets on purpose
C. It only works on local networks
D. It encrypts all segments

الإجابة الصحيحة: A) It does not guarantee segment delivery, orderly delivery, or integrity of the data in segments

ليه صح: لأنه ما بيضمنش أي حاجة: لا وصول الـ segments، لا ترتيبها، لا سلامة بياناتها. تعريف مهم.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.2

IP | Unreliable | Definition | ترجيح: high

.32

Every host has at least one:

ترجمة: كل host عنده على الأقل إيه؟

A. Physical port number
B. DNS resolver
C. Network-layer address (IP address)
D. Transport-layer address only

الإجابة الصحيحة: C) Network-layer address (IP address)

ليه صح: IP address (network-layer address). نقطة مهمة: كل host في الإنترنت عنده IP address. هنفصلها في Chapter 4.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.2

IP address | Hosts | ترجيح: high

.33

The most fundamental responsibility of UDP and TCP is to:

ترجمة: إيه الـ fundamental responsibility بتاع UDP و TCP؟

A. Replace IP with a faster protocol

B. Provide routing within the network core

C. Extend IP's host-to-host delivery service to a process-to-process delivery service

D. Encrypt all application messages

الإجابة الصحيحة: C) Extend IP's host-to-host delivery service to a process-to-process delivery service

ليه صح: يوسّعوا host-to-host delivery بتاع IP لـ process-to-process delivery. ده اسمه transport-layer multiplexing and demultiplexing.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.2

UDP TCP Multiplexing ترجيح: high

.34

Extending host-to-host delivery to process-to-process delivery is called:

ترجمة: توسيع host-to-host delivery لـ process-to-process delivery بيتسمى إيه؟

A. Transport-layer multiplexing and demultiplexing

B. Physical-layer modulation

C. Application-layer caching

D. Network-layer routing

الإجابة الصحيحة: A) Transport-layer multiplexing and demultiplexing

ليه صح: transport-layer multiplexing and demultiplexing. هنقاشه بالتفصيل في القسم اللي after.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.2

Multiplexing Demultiplexing Definition ترجيح: high

.35

UDP and TCP also provide integrity checking by:

ترجمة: UDP و TCP بيوفّروا integrity checking بإيه؟

A. Compressing the segments

B. Encrypting the segments

C. Including error-detection fields in their segments' headers

D. Routing through different paths

الإجابة الصحيحة: C) Including error-detection fields in their segments' headers

ليه صح: بيضيفوا error-detection fields في headers بتاعتهم. ده بتاع الخدمة التانية من الخدمتين الأساسيين.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.2

UDP | TCP | Error detection | Integrity | **ترجيح: high**

.36

How many minimal transport-layer services does UDP provide?

ترجمة: UDP بيوفّر كام خدمة نقل أساسية؟

A. One — only process-to-process data delivery

B. Three — delivery, error checking, and congestion control

C. Two — process-to-process data delivery and error checking

D. Four — delivery, error checking, congestion control, and reliability

الإجابة الصحيحة: C) Two — process-to-process data delivery and error checking

ليه صح: خدمتين فقط: process-to-process data delivery + error checking.
مش بيوفّر reliable transfer ولا congestion control.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.2

UDP | Minimal services | **ترجيح: high**

.37

Like IP, UDP is said to be an unreliable service because:

ترجمة: زي IP، UDP بيتوصف بأنه unreliable ليه؟

A. It always delays packets by 5 seconds

B. It encrypts data so the destination cannot read it

C. It does not guarantee that data sent by one process will arrive intact (or at all) to the destination process

D. It only works on local networks

الإجابة الصحيحة: C) It does not guarantee that data sent by one process will arrive intact (or at all) to the destination process

ليه صح: ما بيضمنش إن البيانات المبعوتة توصل سليمة (أو توصل أصلاً) للـ process المستقبلة. تعريف مهم.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.2

UDP | Unreliable | **ترجيح: high**

.38

Compared to UDP, TCP offers several additional services to applications. Which is one of them?

ترجمة: بالنسبة لـ UDP، TCP بيوفّر خدمات إضافية. إيه واحدة منها؟

A. Smaller segment size

B. Connectionless communication

C. Lower latency than UDP

D. Reliable data transfer

الإجابة الصحيحة: D) Reliable data transfer

ليه صح: reliable data transfer (وأيضاً congestion control). UDP ما بيوفرش أي حاجة من الاتنين.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.2

TCP | Reliable data transfer | **ترجيح: high**

.39

Which techniques does TCP use to ensure data is delivered correctly and in order?

ترجمة: TCP بيستخدم إيه تقنيات عشان يضمن وصول البيانات صح وبالترتيب؟

A. Compression and decompression

B. Routing through different paths

C. Flow control, sequence numbers, acknowledgments, and timers

D. Encryption only

الإجابة الصحيحة: C) Flow control, sequence numbers, acknowledgments, and timers

ليه صح: flow control, sequence numbers, acknowledgments, timers. النقاط دي هنقاشها بالتفصيل في الفصل.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.2

TCP | Reliable data transfer techniques | **ترجيح: high**

.40

TCP converts IP's unreliable service between end systems into:

ترجمة: TCP بيحوّل خدمة IP الـ unreliable بين الـ end systems لـ إيه؟

A. A reliable data transport service between processes

B. An encrypted service only

C. A connectionless service

D. A faster, less reliable service

الإجابة الصحيحة: A) A reliable data transport service between processes

ليه صح: لـ reliable data transport service بين العمليات. ده دور TCP الأساسي فوق IP.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.2

TCP | Reliable data transfer | IP | **ترجيح: high**

.41

In addition to reliable data transfer, what other major service does TCP provide?

ترجمة: بالإضافة لـ reliable data transfer، TCP بيوفّر إيه خدمة تانية؟

A. Encryption of all messages

B. Connectionless communication

C. Faster routing than IP

D. Congestion control

الإجابة الصحيحة: D) Congestion control

ليه صح: congestion control. بيمنع أي TCP connection من إنه يغرق الـ links و الـ routers بـ traffic زيادة.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.2

TCP | Congestion control | **ترجيح: high**

.42

TCP congestion control prevents any one TCP connection from:

ترجمة: TCP congestion control بيمنع أيه؟

A. Sending more than one packet

B. Using UDP as a transport

C. Swamping the links and routers between communicating hosts with an excessive amount of traffic

D. Encrypting all data

الإجابة الصحيحة: C) Swamping the links and routers between communicating hosts with an excessive amount of traffic

ليه صح: يمنع أي TCP connection من إنه يغرق الـ links و الـ routers بـ traffic زيادة. ده لحماية الشبكة كلها.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.2

TCP | Congestion control | **ترجيح: high**

.43

Compared to TCP traffic, UDP traffic is:

ترجمة: بالنسبة لـ TCP traffic، الـ UDP traffic بيتوصف بإيه؟

A. Strictly regulated by TCP

B. Regulated by the routers

C. Always limited to 1 Mbps

D. Unregulated — an application using UDP can send at any rate it pleases, for as long as it pleases

الإجابة الصحيحة: D) Unregulated — an application using UDP can send at any rate it pleases, for as long as it pleases

ليه صح: unregulated. التطبيق يقدر يبعت بأي معدل و لأي مدة من غير ما حد يوقفه. ده فرق أساسي بين TCP و UDP.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.2

UDP | Unregulated | TCP | ترجيح: high

.44

An application using UDP transport can:

ترجمة: التطبيق اللي بيستخدم UDP يقدر يعمل إيه؟

A. Only send for a maximum of 60 seconds

B. Only send during non-peak hours

C. Only send at a rate determined by the routers

D. Send at any rate it pleases, for as long as it pleases

الإجابة الصحيحة: D) Send at any rate it pleases, for as long as it pleases

ليه صح: يبعت بأي معدل و لأي مدة من غير تنظيم. ده ممكن يسبب مشاكل congestion للشبكة — لكنها مشكلة الشبكة مش التطبيق.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.2

UDP | Unregulated traffic | ترجيح: high

.45

Which two minimal services are the only services that UDP provides?

ترجمة: إيه الخدمتين الأساسيين اللي UDP بيوفّرهم وبيقصروا عليهم؟

A. Encryption and authentication

B. Reliable data transfer and congestion control

C. Routing and forwarding

D. Process-to-process data delivery and error checking

الإجابة الصحيحة: D) Process-to-process data delivery and error checking

ليه صح: UDP.process-to-process data delivery + error checking ما بيوفرش reliable transfer ولا congestion control.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.2

UDP | Minimal services | **ترجيح: high**

.46

The transport layer in the Internet is the central piece of the layered network architecture because it:

ترجمة: ليه طبقة النقل بتعتبر central piece في الـ layered architecture؟

A. Manages the physical layer connections

B. Routes packets through the network core

C. Encodes data into bits for transmission

D. Provides communication services directly to application processes running on different hosts

الإجابة الصحيحة: D) Provides communication services directly to application processes running on different hosts

ليه صح: لأنها بتوفّر خدمات اتصال مباشرة لـ application processes على hosts مختلفة. ده دورها المركزي.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1 (intro)

Transport Layer | Architecture | **ترجيح: high**

.47

Which protocol's service model constrains the services that a transport protocol can offer?

ترجمة: بروتوكول إيه اللي بيقيّد خدمات الـ transport؟

A. The physical-layer protocol

B. The underlying network-layer protocol

C. The data-link-layer protocol

D. The application-layer protocol

الإجابة الصحيحة: B) The underlying network-layer protocol

ليه صح: الـ network-layer protocol اللي تحت. لو الـ network ما بيوفرش delay/bandwidth guarantees، الـ transport ما يقدرش يوفرها.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.1

Transport Layer | Network Layer | **ترجيح: high**

.48

When the receiving transport layer processes the received segment, it:

ترجمة: لما طبقة النقل على جانب الاستقبال تعالج الـ segment، بتعمل إيه؟

A. Discards it to save space

B. Makes the data in the segment available to the receiving application

C. Forwards it to another router

D. Encrypts it before delivery

الإجابة الصحيحة: B) Makes the data in the segment available to the receiving application

ليه صح: بتسلم البيانات للتطبيق المستقبل. ده آخر خطوة في رحلة الـ segment من المرسل للمستقبل.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1

Transport Layer | Receiving side | **ترجيح: high**

.49

Which is NOT a service that the IP service model guarantees?

ترجمة: إيه اللي مش خدمة بيضمنها IP؟

A. Orderly delivery of segments

B. All of these are NOT guaranteed by IP

C. Segment delivery

D. Integrity of the data in segments

الإجابة الصحيحة: B) All of these are NOT guaranteed by IP

ليه صح: IP ما بيضمنش أي حاجة من التلاتة. عشان كده اسمه unreliable service.
ملاحظة: "All of these are NOT guaranteed" بمعنى إن الـ IP مش بيضمن أي واحدة منهم — مختلف عن "All of the above" في صياغة سؤال عادي.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.2

IP Unreliable **ترجيح: high**

.50

Which statement best describes the relationship between the transport layer and the network layer?

ترجمة: إيه العبارة الأصح عن علاقة طبقة النقل بطبقة الشبكة؟

A. The two layers are independent and unrelated

B. The network layer extends the transport layer's process-to-process delivery

C. The transport layer replaces the network layer

D. The transport layer extends the host-to-host delivery of the network layer to a process-to-process delivery

الإجابة الصحيحة: D) The transport layer extends the host-to-host delivery of the network layer to a process-to-process delivery

ليه صح: الـ transport بيتوسّع host-to-host بتاع الـ network لـ process-to-process. العلاقة دي hierarchical ومهمة.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.1 + 3.1.2

Transport Layer Network Layer Relationship **ترجيح: high**

.51

The Internet literature (RFCs) refers to the TCP packet as:

ترجمة: أدبيات الإنترنت (RFCs) بتعرف الـ TCP packet بإيه؟

A. A segment

B. A message

C. A datagram

D. A frame

الإجابة الصحيحة: A) A segment

ليه صح: segment. لكن بتقول إن UDP packet ساعات بيتسمى datagram — ده اللي بيخلق لبس.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.2 (terminology)

RFC Terminology Segment **ترجيح: medium**

.52

The Internet literature often refers to the UDP packet as:

ترجمة: أدبيات الإنترنت بتعرف الـ UDP packet ساعات بإيه؟

A. A bit

B. A frame

C. A datagram

D. A segment

الإجابة الصحيحة: C) A datagram

ليه صح: datagram. ده بيخلق لبس لإن نفس الاسم بيتستخدم للـ network-layer packet. عشان كده الكتاب اختار "segment" للاتنين.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.2 (terminology)

RFC Terminology Datagram **ترجيح: medium**

.53

Why does this book refer to both TCP and UDP packets as "segments"?

ترجمة: ليه الكتاب بيسمي TCP و UDP packets بـ "segments"؟

A. To reserve the term "datagram" for the network-layer packet and avoid confusion
B. Because the RFCs require it
C. Because TCP and UDP are the same protocol
D. Because the network layer does not use datagrams

الإجابة الصحيحة: A) To reserve the term "datagram" for the network-layer packet and avoid confusion

ليه صح: عشان يحجز "datagram" للـ network-layer packet ويقلل اللبس.

ملحوظة منهجية مهمة.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.2 (terminology)

Terminology Segment Datagram **ترجيح: medium**

.54

Which layer is responsible for breaking application messages into smaller chunks (if needed) on the sending side?

ترجمة: إيه الطبقة المسؤولة عن تكسير رسائل التطبيق لـ chunks أصغر (لو لزم) على جانب الإرسال؟

A. The data link layer
B. The physical layer
C. The network layer
D. The transport layer

الإجابة الصحيحة: D) The transport layer

ليه صح: طبقة النقل. بـ break (possibly) لـ chunks أصغر وبتضيف header لكل chunk.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1

Transport Layer Sending side **ترجيح: medium**

.55

The book uses the term "datagram" to refer specifically to:

ترجمة: الكتاب بيستخدم "datagram" لـ إيه بالظبط؟

A. A network-layer packet

B. A transport-layer packet (TCP)

C. An application-layer message

D. A transport-layer packet (UDP)

الإجابة الصحيحة: A) A network-layer packet

ليه صح: network-layer packet. عشان يتجنب اللبس مع UDP packet اللي RFCs ساعات بتسميه datagram.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.2 (terminology)

Terminology Datagram **ترجيح: medium**

.56

What is the chapter's discussion of network security (Chapter 8) relevant to in this section?

ترجمة: إيه علاقة Chapter 8 (network security) بالقسم ده؟

A. UDP cannot use encryption

B. TCP requires encryption for all messages

C. IP encrypts all packets by default

D. A transport protocol can use encryption to guarantee that application messages are not read by intruders

الإجابة الصحيحة: D) A transport protocol can use encryption to guarantee that application messages are not read by intruders

ليه صح: بيوضّح إن الـ transport يقدر يستخدم encryption يحمي بيها الرسائل من المتسللين — حتى لو الـ network layer ما بيضمنش confidentiality. ده مثال على إضافة الـ transport لخدمات فوق الـ network.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.1

Security Encryption **ترجيح: medium**

Part 2 — Essay Questions

Essay 1.

Define the role of the transport layer in the layered network architecture, and explain what is meant by "logical communication between processes".

ترجمة: عرّف دور طبقة النقل في الـ layered network architecture، واشرح إيه معنى "logical communication between processes".

The transport layer resides between the application layer and the network layer and is a central piece of the layered network architecture. It has the critical role of providing communication services directly to the application processes running on different hosts.
A transport-layer protocol provides for logical communication between application processes running on different hosts. By logical communication, we mean that from an application's perspective, it is as if the hosts running the processes were directly connected; in reality, the hosts may be on opposite sides of the planet, connected via numerous routers and a wide range of link types.
Application processes use the logical communication provided by the transport layer to send messages to each other, free from the worry of the details of the physical infrastructure used to carry these messages.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1 (intro)

Transport Layer | Logical communication | Definition | **ترجيح: high**

Essay 2.

Explain what the transport layer does on the sending side and on the receiving side of an end system.

ترجمة: اشرح إيه اللي طبقة النقل بتعمله على جانب الإرسال وعلى جانب الاستقبال في الـ end system.

On the sending side, the transport layer converts the application-layer messages it receives from a sending application process into transport-layer packets, known as transport-layer segments in Internet terminology. This is done by (possibly) breaking the application messages into smaller chunks and adding a transport-layer header to each chunk to create the transport-layer segment. The transport layer then passes the segment to the network layer at the sending end system, where the segment is encapsulated within a network-layer packet (a datagram) and sent to the destination.
It's important to note that network routers act only on the network-layer fields of the datagram; they do not examine the fields of the transport-layer segment encapsulated with the datagram.
On the receiving side, the network layer extracts the transport-layer segment from the datagram and passes the segment up to the transport layer. The transport layer then processes the received segment, making the data in the segment available to the receiving application.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1

Transport Layer | Sending side | Receiving side | **ترجيح: high**

Essay 3.

Distinguish between the transport layer and the network layer, in terms of what each provides logical communication between and where each protocol lives.

ترجمة: بيّن الفرق بين طبقة النقل وطبقة الشبكة، من حيث إيه اللي كل واحدة بتوفّر logical communication بينه وفين كل بروتوكول بتعيش.

Whereas a transport-layer protocol provides logical communication between processes running on different hosts, a network-layer protocol provides logical communication between hosts. This distinction is subtle but important.

Transport-layer protocols live in the end systems. Within an end system, a transport protocol moves messages from application processes to the network edge (that is, the network layer) and vice versa, but it doesn't have any say about how the messages are moved within the network core.

In fact, intermediate routers neither act on, nor recognize, any information that the transport layer may have added to the application messages.

The network layer (e.g., IP in the Internet), in contrast, is what gets messages from one host to another across the network core, and routers in the core act on the network-layer fields of the datagram.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.1

Transport vs Network | Distinction | **ترجيح: high**

Essay 4.

Explain how the service model of the underlying network-layer protocol constrains the services that a transport protocol can offer, and give an example of a service that a transport protocol can still offer despite an unreliable network layer.

ترجمة: اشرح إزاي service model بتاع الـ network-layer protocol بيقيّد خدمات الـ transport، واذكر مثال لخدمة الـ transport يقدر يوفّرها رغم إن الـ network مش موثوق.

A computer network may make available multiple transport protocols, with each protocol offering a different service model to applications. The services that a transport protocol can provide are often constrained by the service model of the underlying network-layer protocol.
If the network-layer protocol cannot provide delay or bandwidth guarantees for transport-layer segments sent between hosts, then the transport-layer protocol cannot provide delay or bandwidth guarantees for application messages sent between processes.
Nevertheless, certain services can be offered by a transport protocol even when the underlying network protocol doesn't offer the corresponding service at the network layer:
- A transport protocol can offer reliable data transfer service to an application even when the underlying network protocol is unreliable — that is, even when the network protocol loses, garbles, or duplicates packets. (TCP over IP is exactly this case.)
- A transport protocol can use encryption to guarantee that application messages are not read by intruders, even when the network layer cannot guarantee the confidentiality of transport-layer segments (a topic explored in Chapter 8 on network security).

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.1

Transport Layer | Network Layer | Constraints | Reliable data transfer | **ترجيح: high**

Essay 5.

List and briefly explain the two transport-layer protocols available in the Internet and the services each one provides.

ترجمة: اكتب واشرح باختصار البروتوكولين بتوع النقل اللي الإنترنت بيوفّرهم والخدمات اللي كل واحد بيوفّرها.

The Internet makes two distinct transport-layer protocols available to the application layer:

1. UDP (User Datagram Protocol) — provides an unreliable, connectionless service to the invoking application. UDP provides only two minimal services: process-to-process data delivery and error checking (via error-detection fields in its segment header). Like IP, UDP is an unreliable service — it does not guarantee that data sent by one process will arrive intact (or at all) to the destination process.

2. TCP (Transmission Control Protocol) — provides a reliable, connection-oriented service to the invoking application. TCP offers several additional services beyond what UDP offers:
   - Reliable data transfer: using flow control, sequence numbers, acknowledgments, and timers, TCP ensures that data is delivered from sending process to receiving process, correctly and in order. TCP thus converts IP's unreliable service between end systems into a reliable data transport service between processes.
   - Congestion control: TCP prevents any one TCP connection from swamping the links and routers between communicating hosts with an excessive amount of traffic. UDP traffic, on the other hand, is unregulated — an application using UDP can send at any rate it pleases, for as long as it pleases.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.2

TCP UDP Services Comparison ترجيح: high

Essay 6.

Define the IP service model and explain why IP is called an "unreliable service".

ترجمة: عرّف IP service model واشرح ليه IP بيتسمى "unreliable service".

The Internet's network-layer protocol has a name — IP, for Internet Protocol. IP provides logical communication between hosts.
The IP service model is a best-effort delivery service. This means that IP makes its "best effort" to deliver segments between communicating hosts, but it makes no guarantees.
In particular, IP does not guarantee:
1. Segment delivery (a segment might be lost).
2. Orderly delivery of segments (segments might arrive out of order).
3. The integrity of the data in the segments (a segment's contents might be corrupted).
For these reasons, IP is said to be an unreliable service.
We also mention that every host has at least one network-layer address, a so-called IP address. IP addressing is examined in detail in Chapter 4.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.2

IP | Service model | Best-effort | Unreliable **ترجيح: high**

Essay 7.

Define transport-layer multiplexing and demultiplexing, and explain why it is the most fundamental responsibility of UDP and TCP.

ترجمة: عرّف transport-layer multiplexing و demultiplexing، واشرح ليه ده الـ fundamental responsibility بتاع UDP و TCP.

Extending host-to-host delivery (provided by IP) to process-to-process delivery is called transport-layer multiplexing and demultiplexing.
On the sending side, multiple application processes on a single host can all send messages through the same transport layer; the transport layer multiplexes these messages onto the network layer (each with appropriate port/header information so the receiver can demultiplex them later).
On the receiving side, the transport layer demultiplexes incoming segments: it examines the segment headers to determine which application process should receive each segment, and passes the segment data to that process.
This is the most fundamental responsibility of UDP and TCP because it is what extends IP's host-to-host delivery into a process-to-process delivery service — which is exactly what applications need.
In addition to multiplexing/demultiplexing, UDP and TCP also provide integrity checking by including error-detection fields in their segments' headers. These two minimal transport-layer services — process-to-process data delivery and error checking — are the only two services that UDP provides; TCP adds reliable data transfer and congestion control on top.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.2

Multiplexing Demultiplexing UDP TCP **ترجيح: high**

Essay 8.

Compare the services offered by TCP and UDP. State what each protocol does and does NOT provide.

ترجمة: قارن بين خدمات TCP و UDP. اكتب إيه اللي كل واحد بيوفّره وإيه اللي مش بيوفّره.

Both TCP and UDP extend IP's host-to-host delivery to process-to-process delivery (multiplexing/demultiplexing) and both provide integrity checking via error-detection fields in the segment header. Beyond these two minimal services, the two protocols differ substantially.

UDP:
- Provides: process-to-process data delivery and error checking (only two minimal services).
- Does NOT provide: reliable data transfer, congestion control, connection setup, flow control, sequence numbers, acknowledgments, or timers.
- Like IP, UDP is an unreliable service — it does not guarantee that data sent by one process will arrive intact (or at all) to the destination process.
- UDP traffic is unregulated — an application can send at any rate it pleases, for as long as it pleases.

TCP:
- Provides: process-to-process data delivery, error checking, plus reliable data transfer (using flow control, sequence numbers, acknowledgments, and timers) and congestion control.
- TCP is connection-oriented.
- TCP converts IP's unreliable service between end systems into a reliable data transport service between processes.
- TCP congestion control prevents any one TCP connection from swamping the links and routers between communicating hosts with an excessive amount of traffic.

When designing a network application, the application developer must specify one of these two transport protocols.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.2

TCP UDP Comparison **ترجيح: high**

Essay 9.

Explain what is meant by TCP congestion control and how it differs from UDP traffic in terms of regulation.

ترجمة: اشرح إيه معنى TCP congestion control وإزاي يختلف عن UDP traffic من حيث التنظيم.

TCP provides congestion control. Loosely speaking, TCP congestion control prevents any one TCP connection from swamping the links and routers between communicating hosts with an excessive amount of traffic.
Each TCP connection adjusts its sending rate in response to congestion signals (such as packet loss or delay) observed in the network, so that no single TCP connection unfairly consumes the available bandwidth at the expense of other connections or other traffic. (The exact mechanisms of TCP congestion control are studied in detail later in Chapter 3.)
UDP traffic, on the other hand, is unregulated. An application using UDP transport can send at any rate it pleases, for as long as it pleases. There is no built-in mechanism in UDP to slow down in response to congestion.
This difference has practical consequences: a single UDP application could in principle overwhelm the network with its traffic, while a TCP connection will back off when congestion is detected.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.2

TCP Congestion control UDP **ترجيح: high**

Essay 10.

Why is the transport layer described as a "central piece of the layered network architecture"? Justify with at least three reasons based on the lecture.

ترجمة: ليه طبقة النقل بتعتبر "central piece of the layered network architecture"؟ برّر بثلاث أسباب على الأقل من المحاضرة.

The transport layer is described as a central piece of the layered network architecture for several reasons based on the lecture:

1. Position in the stack: It resides between the application layer and the network layer, acting as the bridge between what applications need and what the network can provide.

2. Direct service to applications: It has the critical role of providing communication services directly to the application processes running on different hosts. From the application's perspective, the transport layer provides logical communication, so processes can exchange messages without worrying about the underlying physical infrastructure.

3. Hides network imperfections from applications: Even when the underlying network layer (IP) is unreliable (loses, garbles, or duplicates packets), a transport protocol such as TCP can offer reliable data transfer to applications — making the network imperfections invisible to the application.

4. Adds new services beyond what the network offers: Beyond reliability, the transport layer can add congestion control (TCP) and even security via encryption (Chapter 8), neither of which is provided by the underlying network layer.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1 (intro)

Transport Layer | Architecture | Role | ترجيح: high

Essay 11.

Describe the journey of an application-layer message from a sending process to a receiving process, mentioning the role of the transport and network layers at each step.

ترجمة: وصّف رحلة application-layer message من process مرسِل لـ process مستقبِل، مع ذكر دور طبقة النقل وطبقة الشبكة في كل خطوة.

1. On the sending side, the sending application process creates an application-layer message and passes it down to the transport layer.
2. The transport layer (possibly) breaks the application message into smaller chunks and adds a transport-layer header to each chunk, creating one or more transport-layer segments.
3. The transport layer passes each segment down to the network layer at the sending end system.
4. The network layer encapsulates each segment within a network-layer packet (a datagram) and sends it toward the destination. Network routers along the path act only on the network-layer fields of the datagram — they do not examine the transport-layer segment fields inside.
5. On the receiving side, the network layer at the destination end system extracts the transport-layer segment from the datagram and passes the segment up to the transport layer.
6. The transport layer on the receiving side processes the received segment (using its header information for demultiplexing and integrity checking) and makes the data in the segment available to the receiving application.

Throughout this journey, the transport layer lives only in the end systems — the routers in the network core never act on or recognize transport-layer information.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1

Transport Layer | Network Layer | Encapsulation | Journey | **ترجيح: high**

Essay 12.

State at least three things that IP does NOT guarantee, and explain the consequence for a transport protocol built on top of IP.

ترجمة: اكتب على الأقل تلات حاجات IP ما بيضمنشهم، واشرح النتيجة على بروتوكول نقل مبني فوق IP.

IP (the Internet's network-layer protocol) is a best-effort delivery service and does NOT guarantee:
1. Segment delivery — a segment might be lost in the network and never arrive at the destination.
2. Orderly delivery of segments — segments might arrive at the destination in a different order than they were sent.
3. The integrity of the data in the segments — the contents of a segment might be corrupted in transit.
Because of these missing guarantees, IP is said to be an unreliable service.
Consequence for a transport protocol built on top of IP:
- If a transport protocol needs to provide reliable data transfer to applications (as TCP does), it must add its own mechanisms to detect lost segments (timers + retransmission), restore order (sequence numbers), and detect corrupted data (error-detection fields + acknowledgments). TCP does all of this using flow control, sequence numbers, acknowledgments, and timers, thereby converting IP's unreliable service into a reliable data transport service between processes.
- If a transport protocol does not need to provide reliability (as UDP does not), it can simply pass through the unreliability of IP, only adding multiplexing/demultiplexing and integrity checking. UDP is therefore also unreliable, like IP.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.1 + 3.1.2

IP Unreliable Consequences TCP UDP **ترجيح: high**

Essay 13.

Explain the terminology used in the Internet literature for transport-layer and network-layer packets, and why this book chooses to refer to both TCP and UDP packets as "segments".

ترجمة: اشرح المصطلحات اللي أدبيات الإنترنت بتستخدمها لـ transport-layer و network-layer packets، وليه الكتاب اختار يسمي TCP و UDP packets بـ "segments".

To simplify terminology, this book refers to the transport-layer packet as a segment. The Internet literature (for example, the RFCs) refers to the transport-layer packet for TCP as a segment but often refers to the packet for UDP as a datagram.
However, this same Internet literature also uses the term datagram for the network-layer packet!
For an introductory book on computer networking such as this, referring to both TCP and UDP packets as datagrams would be confusing, because the term datagram would then refer to two different things: a transport-layer packet (UDP) and a network-layer packet (IP).
Therefore, the book chooses to refer to both TCP and UDP packets as segments, and reserve the term datagram for the network-layer packet. This reduces ambiguity for the reader.

المرجع: Lecture 7 (Ch 3, Lec 1) — Section 3.1.2 (terminology)

Terminology Segment Datagram **ترجيح: medium**